

SUBHRANIL BAUL

AI/ML Engineer | Full Stack Developer

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PROFESSIONAL SUMMARY

Final-year AI/ML engineer at UEM Jaipur (GPA 8.6) with end-to-end experience spanning machine learning, computer vision, full-stack development, and IoT systems working across Python, TensorFlow, OpenCV, React, and Node.js. Published at IEEE IEMECON 2025, co-founded a tech club, and led the technical team for a 3000+ registration hackathon. Proven ability to take projects from prototype to production across edge AI, decentralized applications (Soroban/Stellar), and production-level websites – eager to bring that same execution to a high-impact engineering team.

SKILLS

Programming Languages: Python, JavaScript, Java, C, C++, Rust, SQL

Web Technologies: React, Vite, HTML5, CSS3, Tailwind CSS, Web3, Node.js

AI / Machine Learning: Deep Learning, Computer Vision, NLP, RL, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

Cloud & DevOps: Google Cloud Platform, Firebase, Cloudflare, Git, GitHub, REST APIs, Soroban/Stellar

Hardware & IoT: Raspberry Pi, IoT Systems, Embedded Systems, MATLAB, Vivado, Bluetooth/Gesture Control

Tools & Software: VS Code, Excel, PowerPoint, EmailJS, SUMO (Traffic Simulation), Open Office, MySQL, SQL

Soft Skills: Team Leadership, Problem-Solving, Critical Thinking, Adaptability, Time Management, Public Speaking

EDUCATION

University of Engineering and Management, Jaipur

2023 – 2027 (Expected)

Bachelor of Technology — Computer Science & Engineering (AI/ML Specialization)

GPA: 8.6 / 10.0

Coursework: Machine Learning, Artificial Intelligence, Deep Learning, Reinforcement Learning, Embedded Systems, Computer Vision

WORK EXPERIENCE

Data Analytics Intern – Data Associate L1

July 2025 – September 2025

Infotact Solutions

Remote

- Developed and deployed production ML models using Python and Scikit-learn; applied feature engineering and hyperparameter tuning to achieve 15% accuracy improvement.
- Built end-to-end data pipelines for data ingestion, preprocessing, and transformation to support model training, evaluation, and deployment workflows.

Python Programmer Intern

June 2024 – July 2024

Codsoft

Remote

- Built two client-facing Python projects a fully functional calculator and a Rock-Paper-Scissors game – with error handling and unit tests to ensure production-level code quality, reducing execution time by ~12%.

Freelance Graphic Designer

May 2024 – July 2024

Self-Employed

- Delivered branding and visual design assets for clients including logos, social media creatives, and promotional materials

PROJECTS

AI-Based Face Recognition Attendance System | Python, TensorFlow, OpenCV, Raspberry Pi 5, Firebase

- Developed an edge AI face recognition attendance prototype on Raspberry Pi 5 using TensorFlow and OpenCV, achieving 83% detection accuracy across 20 students and 3 faculty with real-time dashboard analytics.
- Automated attendance tracking for a 23-person cohort, eliminating manual processes and reducing administrative workload.

PactFlow — Decentralized Freelance Escrow Protocol | Soroban (Rust), Stellar Network, React, Firebase, Web3

GitHub Repo: <https://github.com/Dark-97o/PactFlow> | Demo: pact-flow-pi.vercel.app

- Architected and deployed 4 Soroban smart contracts (Escrow, Registry, NFT Marketplace, Reputation) on Stellar Testnet, enabling trustless job matching with automated XLM escrow locking and 95% worker / 5% protocol fee splits.

- Built a full-stack dApp (React + Firebase) with multi-wallet authentication (Freighter, xBull, etc), on-chain reputation scoring, NFT market with 10% royalty engine, and dispute resolution, onboarded 6 real users with live feedback iterations.

Smart Traffic Signal Optimization — Reinforcement Learning | Python, Reinforcement Learning, SUMO

- Designed and built a RL-based adaptive traffic signal control system using Q-Learning and SUMO simulation, dynamically optimized signal phase timings to reduce average intersection wait time by 28% and improve traffic throughput by 18.7%

IoT Smart Wheelchair with Gesture & Bluetooth Control | IoT, Raspberry Pi, Bluetooth, Embedded C

Project Link: <https://devfolio.co/projects/gesture-controlled-wheelchair-5468>

- Engineered an IoT-based smart wheelchair with multi-modal control (gesture, touch, Bluetooth) and autonomous pathfinding, achieving 87.6% gesture recognition accuracy via accelerometer and gyroscope sensor fusion — designed to address mobility challenges for 1M+ physically disabled individuals in India per Census 2011 data.

Acehack Hackathon Website | Vanilla HTML, CSS, JS

GitHub Repo: <https://github.com/AceHack-Hackathon/AceHack-website> | Live : <https://acehack.uem.edu.in/>

- Designed and deployed the official Acehack 5.0 hackathon website supporting 5000+ participant registrations

Pradyog Student Club & ACM UEMJ Websites | React, Vite, Tailwind CSS, Cloudflare, Firebase

Pradyog Live: <https://pradyog.netlify.app/> | ACM UEMJ Live: <https://acm-uemj.uem.edu.in/>

- Built and maintained two club websites serving 300+ active club members

RESEARCH & PUBLICATIONS

IMECON 2025 — 13th International Conference on Intelligent Embedded, MicroElectronics, Communication, and Optical Networks | Paper: <https://ieeexplore.ieee.org/document/11365706>

TensorFlow, Computer Vision, and Edge AI-based Face Recognition Attendance Monitoring System Using Raspberry Pi 5

- Designed and implemented a Raspberry Pi 5-based edge AI attendance system using facial recognition with on-device inference, real-time analytics, and dashboard visualization for automated student and faculty tracking

ACHIEVEMENTS

- 1st Position — Brand Resurrection Challenge | Tech Utopia 2K26, University of Engineering and Management, Jaipur
- 2nd Position — Project Expo 4.0 | Manipal University, Jaipur
- 3rd Position — InnoSpark Event | IEEE Consumer Technology Society, Chandigarh University

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Web-Dev Lead & Tech Lead | Organizer — Acehack 5.0 | Acehack 4.0 Hackathon

January 2025 – Present

- Leading full technical and web infrastructure for UEM Jaipur's flagship hackathon with 3000+ Registrations.
- Managing and supervising a cross-functional team of 40+ members across web, design and tech.

Web Master — ACM UEMJ Student Branch

October 2025 – Present

- Redesigned and maintain the official ACM UEMJ website serving 20+ branch members

Student Vice-Chairman — IEEE APS UEMJ Student Chapter

October 2024 – Present

- Coordinating technical events, workshops, and industry sessions for the IEEE chapter

Co-Founder — PRADYOG Students Club

May 2024 – Present

- Co-founded a student technology and innovation club; grew membership to 30+ students within the first year.
- Organised 10+ IoT based events over 2 years with 200+ students.

Core Team Member — Google Developers Group UEMJ

September 2023 – Present

CERTIFICATIONS

- The Joy of Computing using Python — NPTEL (Elite)
- Data Science with Python — NPTEL (Elite)
- Introduction to Deep Learning & Neural Networks with Keras — Coursera
- Introduction to Data Science in Python — Coursera
- Databases and SQL for Data Science — Coursera
- Smart Manufacturing and Industry 4.0 — IITD-AIA Foundation
- Google Android Developer Virtual Internship – Eduskills AICTE — Cohort 09
- Google AI/ML Virtual Internship – Eduskills AICTE — Cohort 10